

Module 3: Lesson 1

Deep Convolutional Neural Networks: Architectures



Outline

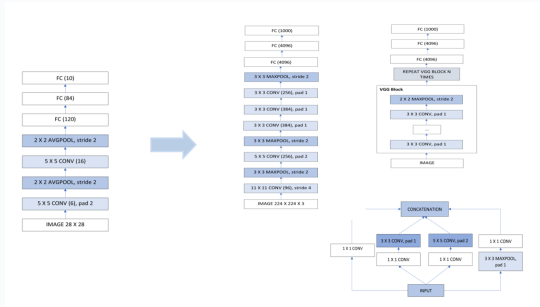
- ▶ LeNet and AlexNet
- ▶ Networks using blocks (VGG) and Network-in-Network (NiN)
- ▶ Multi-Branch networks (GoogLeNet)
- ▶ Residual networks (ResNet, ResNeXt, and DenseNet)
- ▶ From feature engineering to network engineering

CNN Architectures

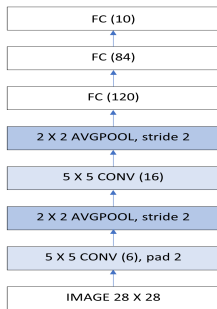
In Module 2, we described all the elements that are necessary to assemble and train a fully operational convolutional neural network.

In Module 3, we are going to delve into additional tools that may enhance the performance of CNNs.

This first lesson describes how new CNN architectures have been proposed and have evolved over time to tackle image recognition tasks.



LeNet



LeNet (LeCun et al., 1998) was one of the first CNNs successfully proposed for image recognition tasks, over other supervised learning tools.

The architecture was designed to process low-resolution images of handwritten digits.

The model combines two layers of 5x5 convolutions with pooling and three fully connected layers.

Originally, it used softmax activations and average pooling.

- LeNet was proposed even before ReLU and max-pooling were “discovered”!

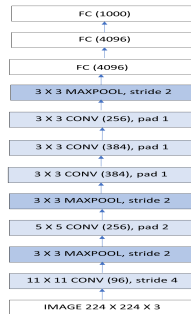
AlexNet

AlexNet (Krizhevsky et al., 2012) can be labeled as the first “modern” CNN.

This network was the first to show that learning algorithms can automatically learn features of images, transcending manually designed features.

New developments in GPU processing reduced the computational burden of deep learning architectures, which sparked their popularity.

AlexNet consists of eight layers, employs ReLU activations, and adds dropout in the fully-connected layers.



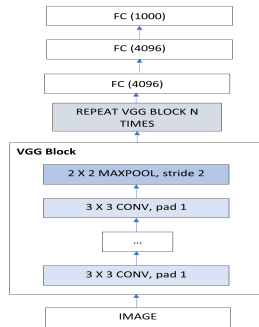
Networks using blocks (VGG)

After AlexNet, the design of neural network architectures has moved to thinking in terms of blocks that repeat patterns of layers.

- The VGG network introduced the preference for deep (many layers) against wide (many channels) networks.

A VGG block (Simonyan & Zisserman, 2014) consists of a sequence of convolutions with 3x3 kernels with padding of 1 (keeping height and width) followed by a 2x2 max-pooling layer with stride of 2 (halving height and width after each block).

The idea is to apply multiple convolutions and avoid a too-quick reduction in the spatial resolution.



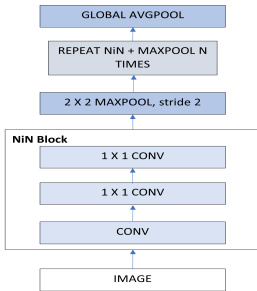
Network-in-Network

One of the major drawbacks of the previous architectures arises from the fully-connected layers that require large amounts of memory.

A Network-in-Network (NiN, Lin et al., 2013) block consists of a typical convolution followed by 1×1 convolutions, which can be thought of as fully connected layers acting independently on each pixel location.

The NiN network gets rid of the giant fully connected layers at the end of the network, which are replaced by a global average-pooling layer.

- This means that the last block must have as many output channels as the number of label classes.



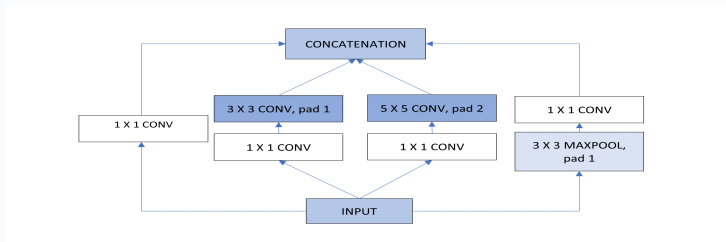
Multi-branch networks (GoogLeNet)

GoogLeNet (Szegedy et al., 2015) combines the strengths of repeated convolutions, NiN, and blocks.

GoogLeNet uses blocks of code (“Inception” blocks) that use four parallel branches.

The Inception blocks explore the image in a variety of filter sizes: Details at different extents can be recognized efficiently by filters of different sizes and different parameters.

- In Lesson 2, we will work on a specific version of this architecture, labeled “Xception.”



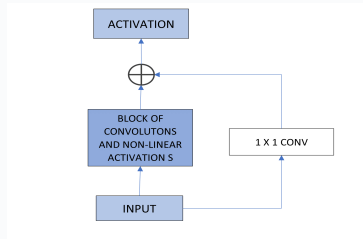
ResNet

When training a neural network, our aim is to approximate a general function $f(\mathbf{x})$ that allows the prediction of the output, given the inputs $\mathbf{x}$.

Similar to a Taylor series expansion, we can obtain an approximation $f(\mathbf{x}) = \mathbf{x} + g(\mathbf{x})$.

The ResNet (Residual Network, He et al., 2016) architecture aims at learning separately the identity mapping, $\mathbf{x}$ and the *residual mapping* $g(\mathbf{x}) = f(\mathbf{x}) - \mathbf{x}$.

In ResNet, a shortcut connection bypasses several convolutional layers, which deal with learning the non-linearities. Both components are then added to obtain the output.



ResNext and DenseNet

ResNet can be understood as a particular case of the Inception block. ResNeXt (Xie et al., 2017) generalizes the ResNet architecture to an Inception block that adopts the same transformation in all the “inside” branches.

At the end of each block of ResNet, the linear and the residual terms are added to obtain an output.

DenseNet, instead, concatenates each output of the residual block, which is fed separately into deeper layers of the network.

Each dense block will increase the number of channels, leading to an excessively complex model. A *transition layer* is used to control this.

From feature engineering to network engineering

Summary of the recent developments in the design of CNNs:

- ▶ AlexNet: Automatic feature extraction with deep CNNs.
- ▶ VGG Networks: Stacking of 3x3 convolutions improve the results of wide layers.
- ▶ NiN: Adding local non-linearities.
- ▶ GoogLeNet, ResNet, ResNeXt: Multi-branch networks that extract different components to approximate general functions.

This has led to new models where the design of the network is also trained on top of the usual model parameters in a neural network: AnyNet.

Summary of Lesson 1

In Lesson 1, we have:

- Described the evolution of CNN architectures over time, from LeNet to ResNext and AnyNet.

⇒ **References for this lesson:**

Chapters 7 & 8 in [Zhang, Aston, et al. *Dive into Deep Learning*. 2022.](#)

TO DO NEXT: This lesson does not have an associated Jupyter Notebook. Please advance to the next lesson.

In the next lesson, we will describe and implement transfer learning using another CNN architecture: Xception.